

Jupiter

Jupiter is a giant among the planets. It is the second-largest body in the solar system after the Sun and the most massive of all the planets. Its visible surface is not solid but the colorful top layer of a deep, thick atmosphere. An extensive family of moons orbits around Jupiter, and a thin faint ring of dust particles encircles it.

FEATURES

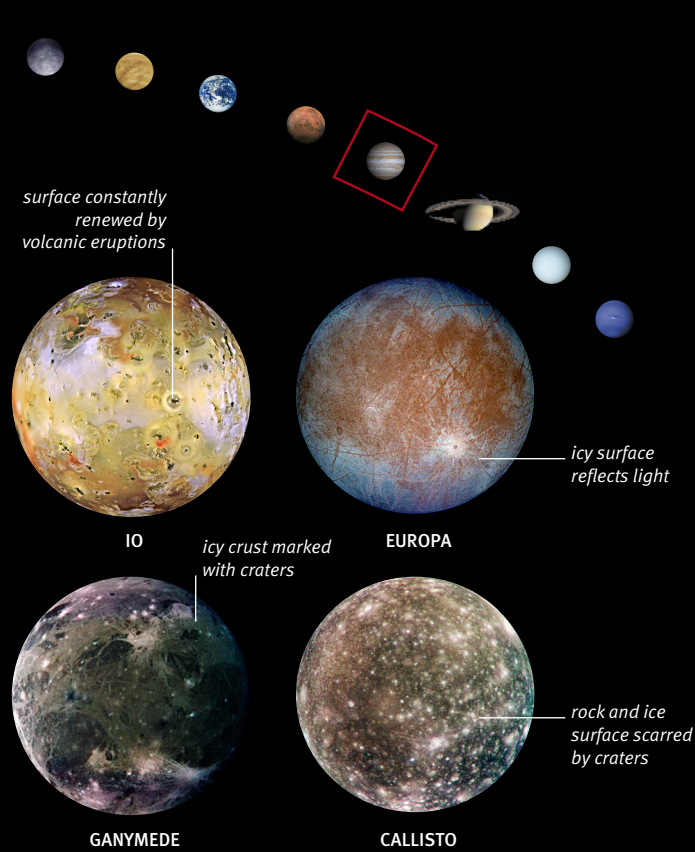
Jupiter is made of hydrogen and, to a lesser extent, helium. Its outer layer is a 600-mile- (1,000-km-) deep hydrogen-rich atmosphere. Underneath this, the planet becomes increasingly dense and hot, and the hydrogen acts like a liquid; deeper still, it acts like a molten metal. At Jupiter's heart is a core of rock, metal, and hydrogen compounds.

The “surface” we see consists of colorful bands of swirling gas. Heat from inside Jupiter, combined with its fast spin, creates the bands and the violent weather within them. The light bands of rising gas and the red-brown ones of falling gas produce winds that give rise to storms and hurricanes. The Great Red Spot is a giant storm that has raged for more than 300 years.



Little Red Spot

In late 2005, astronomers witnessed the birth of Jupiter's second red spot, and second-largest storm—about 70 percent of the size of Earth. It formed between 1998 and 2000 when three white, oval-shaped storms merged.



Galilean moons

Jupiter's four largest moons are collectively known as the Galilean moons. These spherical bodies made of a mix of rock and ice are worlds in their own right. By contrast, most of the other moons are small and irregular in shape.

OBSERVATION

Jupiter shines like a bright silver star in the night sky for about ten months of each year. Its brilliance allows it to stand out from the star background and be recognized. Although its brightness varies, it is never dimmer than Sirius, the brightest star in the night sky.

LOCATING AND LOOKING

Jupiter is best seen when it is close to Earth and on the opposite side of Earth from the Sun (a position known as “opposition”; see p.14). It is then particularly bright because the sunlight that shines fully on it is reflected back by its atmosphere. Its brightness at opposition is at least -2.3 and goes up to a maximum of -2.9 . Jupiter is also in the sky all night long, rising at sunset, being highest in the middle of the night, and setting at sunrise. Opposition occurs every 13 months; dates of oppositions are included in the Special Events listings in the Monthly Sky Guide (pp.44–69).

Jupiter is found within the zodiac band of sky. It takes about 12 months to cross one zodiac constellation before moving into the next. It travels from west to east against the background



of stars, but when at opposition it also goes through a period of retrograde motion (see p.14) and temporarily moves backward in the sky.

For most of the time, Jupiter is the planet with the largest disk size. Surface detail on the planet can be seen through either powerful

binoculars or a telescope. The banded structure, the Great Red Spot, and other cloud features then come into view. A large telescope reveals further spots and structure in the clouds. The slightly squashed appearance of Jupiter's disk, due to its fast spin, is also apparent through a telescope.

Changing view

Our view of Jupiter's surface is constantly changing due to the planet's rapid rotation. Jupiter makes one spin in just under ten hours. This means that features can be seen to move within about ten minutes. These five images show Jupiter over the course of about five hours. The planet spins from left to right, and the Great Red Spot is seen to move as the planet turns. The black dot is the shadow of a Galilean moon.



Naked-eye view

Jupiter's disk shape is clearly visible even to the naked eye. Its shape and brilliance make it easy to identify.



Binocular view

The Galilean moons are visible through binoculars on either side of the planet's equator, changing position each night.



Telescope view

Here, it is possible to discern details: the bands and the Great Red Spot can be seen.